Numerical Analysis homework II

LI YIJIAN 3230102477 *

(Mathematics and Applied Mathematics 2302), Zhejiang University

Due time: October 25, 2025

Abstract

The abstract is not necessary for the theoretical homework, but for the programming project, you are encouraged to write one.

I. Determine the remainder

By Newton interpolation $\forall x \in (x_0, x_1)$:

$$f(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x](x - x_0)(x - x_1)$$

So,we get:

$$\frac{f''(\xi(x))}{2} = f[x_0, x_1, x]$$

i.e.

$$\frac{1}{(\xi(x))^3} = \frac{f[x_0, x_1] - f[x_1, x]}{x_0 - x} = \frac{\frac{f[x_0]-f[x_1]}{x_0-x_1} - \frac{f[x_1]-f[x]}{x_1-x}}{x_0 - x} = \frac{\frac{1}{2x} - \frac{1}{2}}{1 - x} = \frac{1}{2x}$$

so $\xi(x) = (2x)^{\frac{1}{3}}$.

Easily to continuouly extend an easy function like $\xi(x)$ to the clousre to its domain.

So $\xi(x)_{\max} = 4^{\frac{1}{3}}$,$\xi(x)_{\min} = 2^{\frac{1}{3}}$, $f''(\xi(x))_{\max} = \frac{1}{x}_{\max} = \frac{1}{1} = 1$.

II. Interpolation of Polynomials in $\mathbb{P}_{2n}^{+}$

May as well $x_0 < x_2 < \cdots x_n$,we choose

$$\pi_k(x) = \prod_{i=0, i\neq k}^{n} \frac{(x - x_i)^2}{(x_k - x_i)^2} \in \mathbb{P}_{2n}^{+}$$

obviously:

$$\begin{cases} \pi_k(x_i) = 1, i = k \\ \pi_k(x_i) = 0, i \neq k \end{cases}$$

then:

$$p(x) = \sum_{k=0}^{n} f_k \pi_k(x) \in \mathbb{P}_{2n}^{+}, (0 \leq f_k)$$

and $\forall k \in \{0, 1, \cdots n\}$, $p(x_k) = f_k \pi_k(x) = f_k$.

III. The remainder of e^x

III.a

When $n = 1$,LHS $= f[t, t+1] = \frac{e^{t+1}-e^t}{t+1-t} = (e-1)e^t =$ RHS. We suppose $n = k$ has done, then:

$$f[t, t+1, \ldots, t+k+1] = \frac{f[t+1, t+2, \ldots, t+k+1] - f[t, t+1, \ldots, t+k]}{t+k+1-t} = \frac{(e-1)^k(e^{t+1} - e^t)}{(k+1)k!} = RHS$$

*Electronic address: 3230102477@zju.edu.cn

III.b

By III.a we know

$$\frac{(e-1)^n e^0}{n!} = \frac{1}{n!} f^{(n)}(\xi) = \frac{1}{n!} e^{\xi} \Rightarrow \xi = n\ln(e-1) > 0.54n$$

So ξ located to the right of the midpoint $n/2$.

IV. Interpolation of given points

IV.a

By Newton Interpolation, we get $f[0] = 5, f[1] = 3, f[3] = 5, f[4] = 12$.
$f[0,1] = -2, f[1,3] = 1, f[3,4] = 7$.
$f[0,1,3] = 1, f[1,3,4] = 2$
$f[0,1,3,4] = \frac{1}{4}$. So $p_3(f;x) = 5 - 2x + x(x-1) + \frac{1}{4}x(x-1)(x-3) = \frac{1}{4}x^3 - \frac{9}{4}x + 5$.

IV.b

By IV.a,we know

$$p_3'(f;x) = -3 + 2x + \frac{1}{4}[(x-1)(x-3) + x(x-3) + x(x-1)] = -\frac{9}{4} + \frac{3}{4}x^2$$

when $p_3'(f;x) = 0, \Rightarrow x = \sqrt{3}, -\sqrt{3}$, and $p_3''(f;x) = \frac{3}{2}x > 0, (x \in (1,3))$. So, we get $\sqrt{3}$ is a approximate value of $x_{\min}$

V. Interpolation of given points

V.a

$f[0] = 0, f[1] = 1, f[2] = 2^7$
$f[0,1] = 1, f[1,1] = 7, f[1,2] = 2^7 - 1, f[2,2] = 7 * 2^6$
$f[0,1,1] = 6, f[1,1,1] = 21, f[1,1,2] = 2^7 - 8, f[1,2,2] = 5 * 2^6 + 1$
$f[0,1,1,1] = 15, f[1,1,1,2] = 2^7 - 29, f[1,1,2,2] = 3 * 2^6 + 9$
$f[0,1,1,1,2] = 2^6 - 22, f[1,1,1,2,2] = 2^6 + 38$
$f[0,1,1,1,2,2] = 30$

V.b

$f^{(5)}(\xi) = \frac{7!}{2*5!}\xi^2 = 30$, so $\xi^2 = \frac{10}{7}$, so $\xi = \sqrt{\frac{10}{7}} \in (0,2)$

VI. Hermite Interpolation of given points

VI.a

$f[0] = 1, f[1] = 2, f[3] = 0$
$f[0,1] = 1, f[1,1] = -1, f[1,3] = -1, f[3,3] = 0$
$f[0,1,1] = -2, f[1,1,3] = 0, f[1,3,3] = \frac{1}{2}$
$f[0,1,1,3] = \frac{2}{3}, f[1,1,3,3] = \frac{1}{4}$
$f[0,1,1,3,3] = -\frac{5}{36}$
So,$f(2) = 1 + (2-0) - 2(2-0)(2-1) + \frac{2}{3}(2-0)(2-1)^2 - \frac{5}{36}(2-0)(2-1)^2(2-3) = 1 + 2 - 4 + \frac{4}{3} + \frac{5}{18} = \frac{11}{18}$

VI.b

By Hermite Interpolation, we know

$$f(x) - p(f;x) = f[0,1,1,3,3,x]x(x-1)^2(x-3)^2 = \frac{f^{(5)}(\xi(x))}{5!}x(x-1)^2(x-3)^2$$

So the error

$$|f(x) - p(f;x)| \leqslant 2\frac{M}{5!} = \frac{M}{60}$$

VII. Prove equation

We prove by induction, when $k=1$, $\Delta f(x) = hf[x_0, x_1]$, $\nabla f(x) = hf[x_0, x_{-1}]$.
Assume that $k=n$ has been proved, when $k=n+1$:

$$\Delta^{k+1} f(x) = \Delta^k f(x+h) - \Delta^k f(x) = k!h^k f[x_1, x_2, \dots, x_{k+1}] - k!h^k f[x_0, x_2, \dots, x_k]$$

then By Newton Interpolation:

$$\Delta^{k+1} f(x) = (k+1)!h^{k+1} f[x_0, x_1, \dots, x_{k+1}]$$

the same way for ∇ Assume that $k=n$ has been proved, when $k=n+1$:

$$\nabla^{k+1} f(x) = \nabla^k f(x) - \nabla^k f(x-h) = k!h^k f[x_0, x_{-1}, \dots, x_{-k}] - k!h^k f[x_{-1}, x_{-2}, \dots, x_{-k-1}]$$

then By Newton Interpolation:

$$\nabla^{k+1} f(x) = (k+1)!h^{k+1} f[x_0, x_{-1}, \dots, x_{-k-1}]$$

VIII. Prove equation

By defination of $\frac{\partial f}{\partial x_0}$,we know

$$\frac{\partial f[x_0, x_1, \dots, x_n]}{\partial x_0} = \lim_{\Delta x \to 0} \frac{f[x_0 + \Delta x, x_1, \dots, x_n] - f[x_0, x_1, \dots, x_n]}{\Delta x} = \lim_{\Delta x \to 0} f[x_0 + \Delta x, x_0, x_1, \dots, x_n]$$

And f is differentiable at x_0, so f is continuous at x_0, so with finite steps, the error simulate in progress will control by a const c and ε by the continuous at x_0.i.e.

$$|f[x_0 + \Delta x] - f[x_0]| < \varepsilon, |f[x_0 + \Delta x, x_0] - f[x_0, x_0]| < \varepsilon$$

we can prove by induction,every interpolation has only a $c\varepsilon$ difference, in which c is a constant $\Rightarrow$

$$\lim_{\Delta x \to 0} f[x_0 + \Delta x, x_0, x_1, \dots, x_n] = f[x_0, x_0, x_1, \dots, x_n]$$

Besides,we know that the order of interpolation points don't affect the result. So if f is partial differentiable at other variables, we can give $x_0, x_1, \dots, x_n$ a permutation then will get a canonical map to the original question. So the result holds.

IX. Min-Max problem

i.e. to solve:

$$\min_{a_i \in \mathbb{R}, i \in 1,2,\cdots n} \max_{x \in [a,b]} |a_0 x^n + a_1 x^{n-1} + \dots + a_n| = |a_0| \min_{a_i \in \mathbb{R}, i \in 1,2,\cdots n} \max_{x \in [a,b]} |x^n + \frac{a_1}{a_0} x^{n-1} + \dots + \frac{a_n}{a_0}|$$

$\Rightarrow$

$$|a_0| \min_{a_i \in \mathbb{R}, i \in 1,2,\cdots n} \max_{x \in [a,b]} |x^n + a_1 x^{n-1} + \dots + a_n| = |a_0| \min_{a_i \in \mathbb{R}, i \in 1,2,\cdots n} \max_{x \in [-\frac{b-a}{2}, \frac{b-a}{2}]} |x^n + a_1 x^{n-1} + \dots + a_n|$$

Notice that $a < b$ otherwise the value is trivial $\Rightarrow$

$$|a_0| \min_{a_i \in \mathbb{R}, i \in 1,2,\cdots n} \max_{x \in [-1,1]} |(\frac{b-a}{2})^n x^n + (\frac{b-a}{2})^{n-1} a_1 x^{n-1} + \dots + a_n| = (\frac{b-a}{2})^n |a_0| \min_{a_i \in \mathbb{R}, i \in 1,2,\cdots n} \max_{x \in [-1,1]} |x^n + a_1 x^{n-1} + \dots + a_n|$$

$$\geqslant (\frac{b-a}{2})^n |a_0| \min_{a_i \in \mathbb{R}, i \in 1,2,\cdots n} \frac{1}{2^{n-1}} = |a_0| \frac{(b-a)^n}{2^{2n-1}}$$

In which the last "$\geqslant$" is by the corollary in the book. And we know that choose Chebyshev Polynomials can get the equal.

X.The minimal element in infty norm

If NOT, we assume $\exists p \in \mathbb{P}_n^a$. s.t.

$$||p||_\infty < ||\hat{p}_n||_\infty, i.e. \quad \max_{x\in[-1,1]} |p(x)| < \frac{1}{|T_n(a)|}$$

Let $Q(x) = \hat{p}_n(x) - p(x)$, then in the $n+1$ extreme points ($\in [-1,1]$) of $T_n(x)$,denote by $x_0, x_1, \dots, x_n$

$$Q(x_k) = \frac{(-1)^k}{T_n(a)} - p(x_k), k = 0, 1, \dots, n$$

just like the proof in Chebyshev Theorem $Q(x)$ will change the signals $n+1$ times, so $Q(x)$ have n zeros in $[-1,1]$. Meanwhile, $Q(a) = 1 - 1 = 0$.So $Q(x)$ have $n+1$ zeros, and $Q(x) \in \mathbb{P}_n$,so $Q(x) \equiv 0$. Contracdiction!

XI.Lemma Proof

$$RHS = \binom{n}{k} t^k (1-t)^{n-k} \frac{n-k}{n} + \binom{n}{k+1} t^{k+1} (1-t)^{n-k-1} \frac{k+1}{n}$$

$$LHS = \binom{n-1}{k} t^k (1-t)^{n-1-k}$$

Notice that $\binom{n}{k} \frac{n-k}{n} = \frac{n!}{k!(n-k)!} \frac{n-k}{n} = \frac{n!}{(k+1)!(n-k-1)!} \frac{k+1}{n} = \binom{n}{k+1} \frac{k+1}{n} = \frac{(n-1)!}{k!(n-k-1)!} = \binom{n-1}{k}$. So

$$RHS = \binom{n-1}{k} t^k (1-t)^{n-k-1} (1-t+t) = \binom{n-1}{k} t^k (1-t)^{n-1-k} = LHS$$

XII.Lemma Proof

$$\int_0^1 b_{n,k}(t)dt = \int_0^1 \binom{n}{k} t^k (1-t)^{n-k} dt = \binom{n}{k} B(k+1, n+1-k) = \binom{n}{k} \frac{\Gamma(k+1)\Gamma(n+1-k)}{\Gamma(n+2)} = \frac{n!k!(n-k)!}{(n-k)!k!(n+1)!} = \frac{1}{n+1}$$